

Machine Vision Final Project Report

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Executive Summary

Here are what I consider to be the most valuable learnings from over the course of this project.

- Trying to train directly on raw videos produced poor results.
 - A two-stage CNN suffered from chronic vanishing gradients and failed to learn.
 - After many small fix attempts, a ResNet eventually fixed the vanishing gradients problem, but trained very slowly, and with underwhelming final accuracy.
- If you can justifiably change your regression problem into a classification problem, then the learning task likely becomes much easier.
- Preprocessing videos using pose estimation immediately gave us information-rich representations that quickly achieved better results, compared to learning features from scratch.
- Augmentating the data helped achieve a sufficiently large, class-balanced training set, and subtle smoothing helped avoid confusing the model with sharp movements.
- There is an art to balancing model size/complexity, weight decay, and learning rate scheduling to learn sufficiently complex relationships, avoid overfitting, and not-waste training time.

Learning About the Data

To get a sense of our training data, I watched most of our training videos and inspected some metadata:

- All videos seem to be shot with 24 frames-per-second, for a total length of roughly 90 to 360 frames (i.e. roughly 3.5 to 10 seconds).
- Most videos are shot in wide-screen, meaning if we squish videos to a 224x224 square resolution for training, the subject will look squished. Hopefully the push-up movement would still be apparent.
- Videos are shot from all angles (from the side, from the front, from above), and sometimes with camera movement. Any feature learning we do will have to take into account how the camera affects the subject's position in plane-space.
- Since pushups are relatively smooth movements, we may be able to stride over frames (i.e. for each video, only look at every 5th or 10th frame) during both training and inference and still see the subject's movement. This way, we could save a lot of memory overhead.

1) Learn the Features from Scratch (With Underwhelming Results)

I was inspired to see whether the task was possible without worrying too much about feature engineering. The promise of a deep learning approach is that the model may be able to learn useful data representations on its own, and I was excited to see what could be done by using raw videos.

In hindsight: all of these approaches suffered from severe problems:

- Underwhelming final training accuracy
- very long training time (30 epochs could easily take 3 hours).
- many epochs without improvement
- memory overload: training would consistently take up anywhere from 10GB to 14GB of unified MPS memory (on Mac)

In the end, I defaulted to pose estimation, as a way of working with significantly smaller data that was still informative to our task. But first, below is a record of what I tried:

1.1): A Basic CNN for both space and time

To track the subject's pushup movement over space and time, I proposed a two-part architecture:

- A series of 2D-convolutions on each frame, to extract spatial features about the subject and its surroundings.
- A series of 1D-convolutions across frames, to see how those spatial features changed over time.
- A linear head.

I conducted training with the following fixed parameters:

- 10 epochs
- mean squared error loss (i.e. regression)
- a stride of every 10th frame when loading training data, with max frames set to 100.
- learning rate = $1e-3$

Below is the summary of the first model, which I then modified a number of ways in different combinations:

Stage	Input Shape	Operations	Output Shape
Spatial Encoder	$(B \times T, 3, 224, 224)$	6x "Conv2d (Stride 2) + ReLU" + Linear(640, 2)	$(B \times T, 2)$
...	$(B \times T, 2)$	expand and transpose	$(B, 2, T)$
Temporal Conv	$(B, 2, T)$	3x Conv1d (Stride 2) + ReLU	$(B, 18, 13)$
...	$(B, 18, 13)$	Flatten	$(B, 234)$
Classifier	$(B, 234)$	Linear	$(B, 1)$

Table 1: VideoConvModel: 27,659 parameters

Disclaimer

I reached for a CNN before even trying a fully-connected MLP.

An MLP has much weaker inductive bias than a CNN, since the MLP does not presume translational equivariance over the input data. This is helpful sometimes in that an MLP could potentially learn far more complicated relationships between inputs and outputs, but at the cost of much larger memory

overhead. However, the CNN would have the advantage of learning to understand pushups regardless of where the subject lives in the frame, whereas an MLP would have to relearn this whenever the subject appears in a new locatin. The weak inductive bias also makes the MLP more likely to overfit. For this reason, along with our memory footprint limits, a CNN seemed preferable. In hindsight, starting with an MLP would have been helpful to get a baseline understanding for how difficult the our learning problem is. If even a relatively-deep MLP struggles to acheive high training accuracy, then our CNN would likely be insufficient. Luckily, this turned out to not be necessary.

The Result: Vanishing Gradients

Every attempt to use a CNN for building spatial features resulted in the model only ever predicting the mean label for each batch.

Here are the results over 10 epochs of training for a sample run:

Epoch	Validation Loss	Validation Accuracy	St.Dev. of Predictions	Norm of Weights
1	11.6948	0.00	0.00	7.687
2	1.8193	0.60	0.00	7.785
3	1.4695	0.13	0.00	7.804
4	1.7752	0.60	0.00	7.792
5	1.4428	0.60	0.00	7.806
6	1.6283	0.60	0.00	7.801
7	2.1680	0.60	0.00	7.798
8	1.7379	0.60	0.00	7.811
9	1.4397	0.60	0.00	7.830
10	1.4846	0.60	0.00	7.837

Table 2: Training Results

Some observations:

- A prediction standard deviation of 0.00 for every validation pass is a sign that the model has learned to simply predict the mean of the validation set.
- Validation loss and the total euclidean norm of the model weights are changing (minimally) per pass. So, the model is technically “learning” something, but not nearly enough to affect predictions (as shown by the st.dev.s)
- Validation accuracy is capped at 60% because the mean
- These 10 epochs took 10 minutes to run, and the model was consuming about 6.54GB VRAM on my Macbook’s MPS the whole time, i.e. painful.

I tried modifying the CNN a number of ways:

- added Batch Normalization to each convolutional block in both spatial and temporal stages.
- less aggressive compression in Linear bottlenecks (adjusting the convolutions’ input/output shape accordingly).
- Increased temporal convolutions kernel size from the default 3 to 5, to try to capture more push-up movement in each step.

Each tweaked model varied significantly in size, up to 669,569 parameters total.

Across 5 attempts with different adjustments, results were similar. The model did not effectively train and only ever predicted the mean of the validation set.

1.2): ResNet for Spatial Features

After several failed attempts, I thought a small ResNet with skip connections might do solve the vanishing gradients problem. The skip connections make the loss landscape smoother, which makes calculating gradients easier.

So I replaced the spatial feature stage with a pre-built ResNet18, beginning with all weights frozen to avoid the computational overhead of computing so many more gradients.

Stage	Input Shape	Operations	Output Shape
Spatial Features	$(B \times T, 3, 224, 224)$	Pre-trained ResNet18 (Frozen)	$(B \times T, 512, 1, 1)$
...	$(B \times T, 512, 1, 1)$	Tranpose	$(B, 512, T)$
Temporal Processor	$(B, 512, T)$	3x [Conv1d + BatchNorm + LeakyReLU]	$(B, 32, 13)$
...	$(B, 32, 13)$	Flatten	$(B, 416)$
Classifier	$(B, 416)$	Linear	$(B, 1)$

Table 3: VideoResNetModel: 11,408,682 parameters

Training parameters were the same as before: * 10 epochs * mean squared error loss (i.e. regression) * a stride of every 10th frame when loading training data, with max frames set to 100. * learning rate = $1e-3$

This architecture successfully trained:

- **St.Dev.s of Predictions:** $\{.63, .94, .62, .94, .91, .90, 1.28, .86, .80, .88, .90\}$
- **Final epoch validation loss:** 1.015
- **Final euclidean norm of weights:** 17.615

Training Load: 20 minutes (on macbook MPS), , 12.GB VRAM consumed

Full training set evaluation: 63% accuracy.

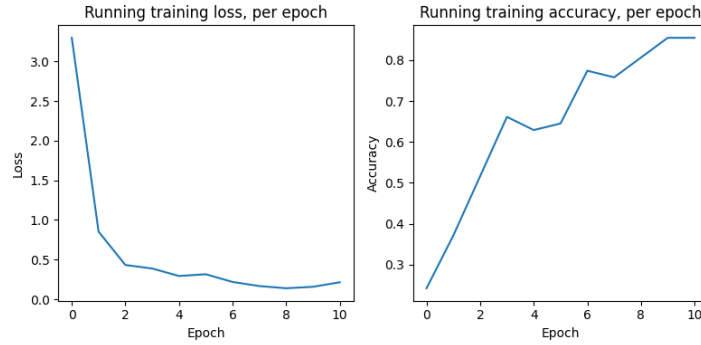


Figure 1: Training loss and accuracy, per epoch

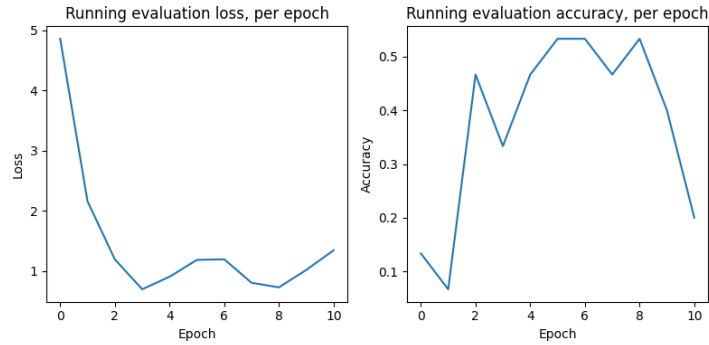


Figure 2: Validation loss and accuracy, per epoch

1.2) Building upon ResNet

I tried a series of modifications to see if this ResNet backbone could successfully tackle the problem:

1.2.0) Unfreeze some ResNet weight

ResNet18 is pre-trained on ImageNet, and thus understands a lot about various objects, including people and landscapes. However, it may not understand anything specific about pushups. So, I unfroze the final layer of the resnet backbone, to see if weight-updates would help the ResNet better recognize spatial features important for counting pushups.

Result: - **Full training set evaluation:** 43% accuracy

I refroze all ResNet-backbone weights going forward.

1.2.1) Switching to Classification

In previous attempts, the model would be penalized for even small deviations from integer predictions, and in the end, the rounded, predicted number of pushups was often one off from the actual value, suggesting the model was getting close but not within rounding distance.

I tried switching to ten-class classification to force integer predictions, using cross-entropy-loss. I amended the output layer of the Resnet Hybrid architecture above to output with shape [Batch, 10], and predicted with the max of these raw scores.

Result: - **Full training set evaluation:** 65% accuracy after ~20 epochs (1.5 hours)

Intuitively, classification should be slightly easier - it's like asking the model to pick amongst 10 buckets, instead of asking it to hit an exact spot on a continuous line.

All model attempts going forward used cross-entropy loss, instead.

1.2.2) Trying Data Transformations

I was concerned about the model being inhibited by how small our dataset is. If I could implement similar transformations to those in Practical 9, then our model would see slightly different versions of training points during each epoch, which would artificially “expand” the size of our dataset and hopefully nudge the model to learn real push-up movement patterns instead of memorizing coincidental video details.

Online research suggested it was very important that, within a given video, transformations needed to be applied the same way to all frames. Random cropping, angle shifts, or color jitter applied non-uniformly would make the video look chaotic and warbly, making the learning problem much more difficult. So, I built wrapper functions which generated a random seed per video and guaranteed the same random operations were copied across all frames in the same video.

I tried two different sets of training and validation transformations: one “difficult” and one “easier” (written below).

Difficult

```
## Directly copied from Practical 9
train_transform = T.Compose([

    T.Resize( (224, 224) ),
    T.RandomCrop(150),

    T.RandomHorizontalFlip(p = .1),
    T.RandomVerticalFlip(p = .1),
    T.RandomRotation(degrees = 10),

    T.ColorJitter(brightness=0.1, contrast=0.1, saturation=0.1),

    T.Normalize(mean = [ 0.485, 0.456, 0.406 ],std = [ 0.229, 0.224, 0.225 ])
])

val_transform = T.Compose([

    T.Resize( (224, 224) ),
    T.CenterCrop(150),
    T.Normalize(mean = [ 0.485, 0.456, 0.406 ],std = [ 0.229, 0.224, 0.225 ])
])
```

Easier

```
train_transform = T.Compose([

    # --- random crop is confined to the edges
    T.Resize((240, 240)),
    T.RandomCrop(224),

    # --- Removed vertical flip, so all pushups are still going downwards
    T.RandomHorizontalFlip(p=0.5),

    # --- Reduced from 10 to 5 degrees
    T.RandomRotation(degrees=5),
```

```

# --- smaller brightness and contrast jitter, no saturation change
T.ColorJitter(brightness=0.05, contrast=0.05),

T.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

val_transform = T.Compose([

    # removed random crop
    T.Resize((224, 224)),

    T.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

```

In the end, both implementations made the model significantly worse, and the model required significantly more time to train up to these values (~60 epochs):

Results:

- **Difficult:** 36% final training accuracy
- **Easier:** 55% final training accuracy

The model was not learning enough about push-ups for added “push-up variation” to be helpful. It was just making the classification problem harder.

The RandomCrop(150) and VerticalFlip were the likely killers in the difficult transformations.

- Given we’re already resizing to (224, 224), the subject is already scrunched as is. A random resize to (150, 150) might cut out entire sections of the subject’s body.
- Pushups always happen downward. Asking the model to learn to recognize pushups “upward” was, in hindsight, unfair.

I scrapped both transformation sets going forward.

(Later, different transformations became very useful - more on that in section 2.0.1 below).

1.2.3) Trying Transformer Architecture for Temporal Stage

Maybe the problem was that convolutions weren’t strong enough to track the spatial features’ movement over time. Given a video is naturally a sequence of frames, I imagined a small transformer for our temporal stage might improve things. The model could learn to attend to certain important frames in the sequence (like, perhaps, the beginning, when the subject is in the start position, to use as reference for determining when a pushup was completed).

Result: 25% final accuracy (~20 epochs)

The transformer did not help at all. After more online research on transformers, I concluded this this architecture would likely need significantly more data to learn anything valuable.

1.2.4) The same temporal convolutions, but deeper.

The next unlock came from simply adding another layer each to the temporal and classification stages of the original resnet architecture:

After 25 epochs, this architecture was able to achieve 100% during some training epochs, with a final validation accuracy of 75%. It was clearly overfitting, but it was reassuring to see that an architecture like this could at least learn something.

Stage	Input Shape	Operations	Output Shape
Spatial Features	$(B \times T, 3, 224, 224)$	Pre-trained ResNet18 (Frozen)	$(B \times T, 512, 1, 1)$
...	$(B \times T, 512, 1, 1)$	Unpack dimensions	$(B, T, 512)$
Sequence Prep	$(B, T, 512)$	Concat CLS Token + Pos. Encoding	$(B, T + 1, 512)$
Transformer Encoder	$(B, T + 1, 512)$	2 Layers [4 Attn. Heads + Feed Forward]	$(B, T + 1, 512)$
...	$(B, T + 1, 512)$	extract CLS value in the front	$(B, 512)$
Classifier	$(B, 512)$	Linear	$(B, 10)$

Table 4: VideoTransformerModel: 15,439,434 parameters

Stage	Input Shape	Operations	Output Shape
Spatial Features	$(B \times T, 3, H, W)$	Pre-trained ResNet18 (Frozen)	$(B \times T, 512, 1, 1)$
...	$(B \times T, 512, 1, 1)$	Unpack dimensions	$(B, T, 512)$
Temporal Processor	$(B, 512, T)$	4x [Conv1d + BatchNorm + LeakyReLU]	$(B, 32, 7)$
...	$(B, 32, 7)$	Flatten	$(B, 32 \times 7)$
Classifier	$(B, 32 \times 7)$	2x [Linear + LeakyReLU]	$(B, 10)$

Table 5: VideoResNetModel: 11,726,522 parameters

1.2.5) Data Augmentation for Class Balance

It was at this phase I recalled the importance of class balance - it had not dawned on me until now that nearly all training videos contained 2, 3, and 4 pushups. Given we don't know the class balance of our test set, I needed the model to reliably predict anywhere from 1 to 10, so I needed a way of expanding the training set to learn higher pushup counts.

After some online research, I landed on the following augmentations:

Operation	Labels Gained
Cut all 2's in half	1's
Cut all 4's in half	2's
Extend 3's by about 2/3	5's
Double 3's	6's
Extend 4's by about 3/4	7's
Double 4's	8's
Triple 3's	9's
Double 5's (including previously-generated)	10's

Table 6: Data Stretching and Slicing Operations

In the end, we achieved a more balanced dataset, with the following label counts:

Slicing needed to be done with care to avoid jump cuts, which could be misinterpreted by the model as pushups. To keep transitions smooth, I concatenated "palindrome stitches", i.e. the *reverse* list of frames with the original, like so:

- Doubles = Original + Palindrome
- Triples = Original + Palindrome + Original
- 3 to 5 = Original + (Palindrome * 66%)

Label	1	2	3	4	5	6	7	8	9	10
Count	33	58	34	21	36	36	22	21	34	34

Table 7: (1.2.5) training set labels and counts (329 total)

- 4 to 7 = Original + (Palindrome * 80%)

I used 66% and 80% instead of 60% and 75% stitches for the 5's and 7's because many videos contained a little bit of "dead space" at the beginning before the subject would first move, which would downstream lead the 5th and 7th push-ups to get cut off too early. The extra stitching helped ensure the final pushup was completed.

All cut-in-half videos were cut at the nearest integer to the midpoint frame.

The result was still bad: the ResNet Hybrid from (1.2.4) achieved just 19% training accuracy after 29 epochs

i Back to Regression after Data Augmentation

With a more balanced training set, I wondered whether regression would be more appropriate again. Perhaps earlier attempts were hamstrung by our training set's low variance. The answer is emphatically, "no:"

- After 30 epochs of training, ResNet Hybrid (1.2.4) achieved its best weights on epoch 3, despite a learning rate scheduler lowering the learning rate to 3e-7 by epoch 25.
- This best model achieved just 6% accuracy.
- Some predictions were wildly high (two videos were predicted to contain 14 pushups, and another 16).

Lesson learned: regression seems generally harder for this problem.

1.2.6) Simpler Augmentation, Just for Training set Size

As an experiment, I made a copy of every video by first flipping it horizontally and reversing the order of frames. This doubled the size of the training set, but retained the harsh class imbalance.

Unsurprisingly, ResNet Hybrid (1.2.4) achieved 75% accuracy on this set, this time after just 9 epochs. This mirrors the result from (1.2.4), and we can chalk up the faster convergence to the dataset being larger.

i Training Hygiene

Over the course of the failures throughout (1), I stumbled upon a number of helpful training tips to make life easier:

- **weight decay:** After each batch, the typical weight update is dictated by the learning rate. But to keep the model from overfitting, we can multiplicatively pull the model towards smaller updates using a small weight-decay. I tinkered with values of 1e-4 to 9e-4, with varying degrees of success.
- **A learning rate scheduler:** If the learning rate is too high, the weight updates may be too large, which can keep us from descending towards a minimum in the loss landscape. After a few epochs of no improvements, you can "schedule" your learning rate to drop by some percentage, making it more likely that an update will lead you downhill. I tinkered with schedule dops of 1/2 to 1/10th the previous learning rate after 2 or 3 epochs of no improvement in validation loss.
- **AdamW:** The AdamW optimizer is specifically designed to decouple learning rates and weight decay (REVIEW THIS).

- **Early stopping:** I eventually implemented a rule where the training loop would end early if the learning rate ever fell below $1e-8$. At that rate, weight updates are unlikely to be meaningful.
- **Saving your Best:** You don't necessarily have to keep the model returned after the final epoch. Sometimes, the weights associated with lowest validation loss (and thus the one which in theory will generalize best) will happen sometime in the middle of training. I eventually implemented a section of my training protocol which, anytime a new "best validation loss" was achieved would save a "checkpoint" of the model state, optimizer state, scheduler state, and epoch when the loss occurred. That way, I always had access to the best model "so far."

I implemented each of these in my final, successful attempts (2).

2) Use Post Estimation to Follow the Whole Person

While it is true that the subject's relationship with its surroundings (like the ground) changes over the course of a push-up, the essence of a push-up is mostly about the body's joints' orientation to each other:

- the shoulders and wrists coming together as the elbow moves backwards,
- the torso and back lowering and raising
- the feet staying fixed

Tracking these movements alone may be enough for our task. This approach also allows us to use significantly lower-dimensional data for every frame, which speeds up training.

2.0.1) The Same Data Augmentation from (1.2.5)

Our efforts with the ResNet were not all a waste! We can achieve a class-balanced training set by applying the same transformations as in (1.2.5), this time on pose estimations.

As before, we end up with 329 total training "videos" (pose tensors) distributed roughly evenly over labels 1 through 10.

2.1) First Attempt

I tried returning to our roots: a CNN + Linear Classifier, this time after pre-processing videos through MediaPipe. This model could quickly train 100 epochs on our 329 preprocessed pose files in about a minute.

Stage	Input Shape	Operations	Output Shape
Preprocess Videos	(B, C, T, H, W)	Mediapipe Joint Coordinate Estimation	$(B, T, 33, 4)$
...	$(B, T, 33, 4)$	Flatten and transpose	$(B, 132, T)$
Temporal Processor	$(B, 132, T)$	2x [Conv1d + BatchNorm + LeakyReLU]	$(B, 256, 1)$
Classifier	$(B, 256, 1)$	Flatten + Linear + LeakyReLU + Linear	$(B, 10)$

Table 8: PoseModel: 179,482 parameters

Training accuracy: 72% on our expanded class-balanced training set, 61% on the original training set

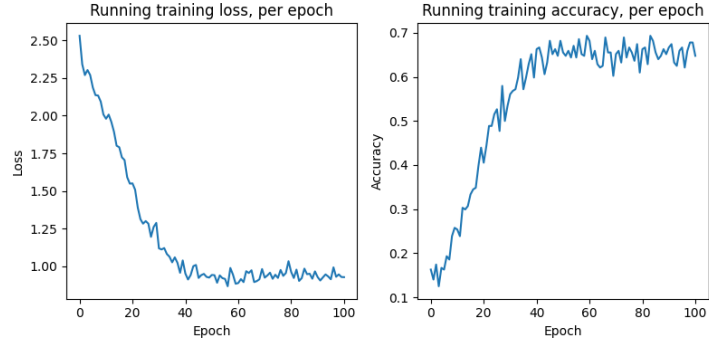


Figure 3: Training loss and accuracy, per epoch

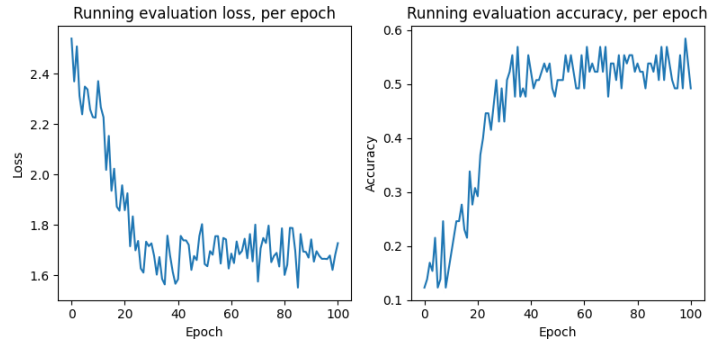


Figure 4: Validation loss and accuracy, per epoch

Our confusion matrix shows some peculiar behavior.

- Most mistakes are at the margin, predicting one up or down of the actual value (which is good).
- The model has a tendency to predict 6 on high-count videos. 6 is relatively highly represented in the dataset, but not by much.
- There are few instances of gross misses (e.g. confusing 8 for 2, or 2 for 7, etc.) These don't seem to be comprehensively explained by our data augmentation (if it were, we would expect more confusions between 3 and 5, or 4 and 7).

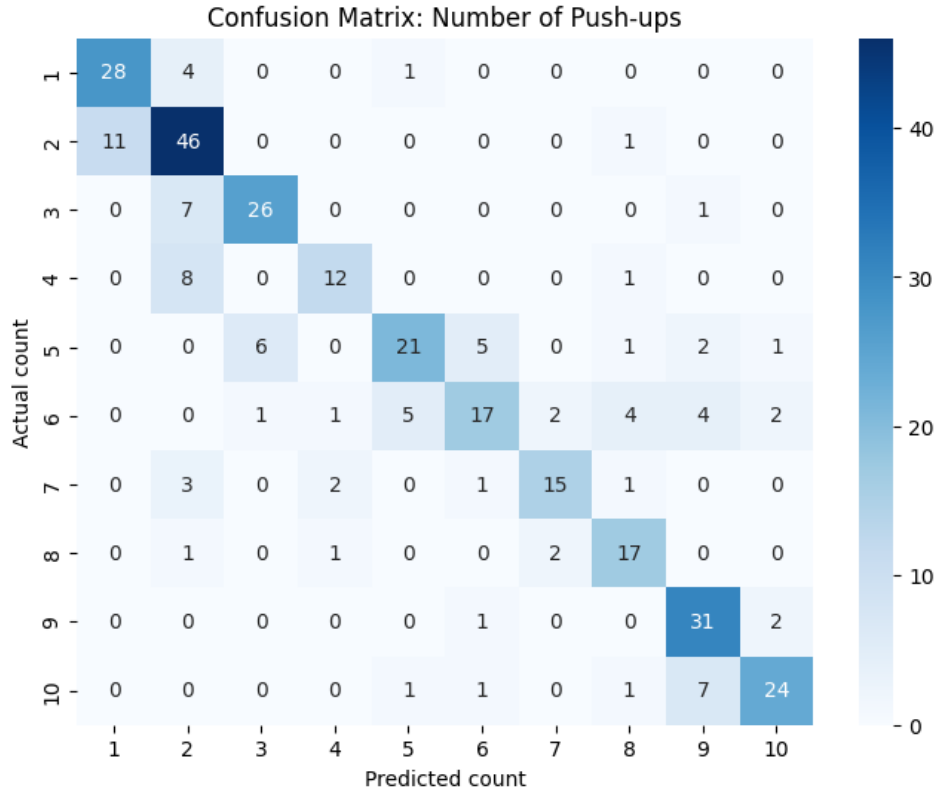


Figure 5: Confusion matrix, predicted vs. actual push-up counts

2.1.1) Small Modifications

Now that the time cost of training was so low, I could play with several tweaks:

- Adding small FC layer before temporal convolutions
 - it became helpful to allow the model to learn a more compressed representation of the joint coordinates before passing this representation to the temporal convolutions.
 - Expanded representations (i.e. going from 132 joint*features to 256) also helped, though less so than compression.
 - Adding an FC layer with the same in and out channels made the model worse.
 - * **Takeaway:** Layers with the same input and output channels can be dangerous. The model is liable to learn non-useful representations, like rotation, translation, or simply noise.
- augmenting frame stride and max frame length (holding architecture constant)
 - stride 10, max frame length 100: 93.5% accuracy
 - stride 5, max frame length 200: 89% accuracy
 - stride 3, max frame length 300: 72% accuracy

- stride 10, max frame length 200: 7% accuracy
- **Takeaway:** Since all videos are going to be under 1000 frames, ensuring $stride \times maxlength = 1000$ guarantees we don't cut off videos too early. The longest video in our class-balanced set was FRAME COUNT. For videos at this speed, a stride of 10 seems to capture enough of the push-up movement to be meaningful. Lower strides run the risk of capturing glitchy movement, which might inflate the pushup count prediction. And if $stride \times maxlength \ll 1000$, then we end up padding the video with too much “stand-still” frames, which wash out the signal once our temporal layer hits the Global Average Pool at the end.
- weight decay
 - choosing the right weight decay called for balancing with the model architecture. A slightly larger model would sometimes overfit quickly, and the learning rate scheduler would sometimes exit the training loop after far fewer than 100 epochs. With these models, increasing weight decay had the dual advantage of tempering overfitting while allowing the model to train for longer before hitting the learning rate limit.

2.2) Second Attempt

Incorporating the above edits, we get an architecture that looks like this:

Stage	Input Shape	Operations	Output Shape
Preprocess Videos	(B, C, T, H, W)	Mediapipe Joint Coordinate Estimation	$(B, T, 33, 4)$
...	$(B, T, 33, 4)$	Flatten	$(B, T, 33 \times 4)$
Spatial Embedding (FC)	$(B, T, 33 \times 4)$	Linear + LeakyReLU	$(B, T, 32)$
...	$(B, T, 32)$	transpose	$(B, 32, T)$
Temporal Processor	$(B, 32, T)$	3x [Conv1d + BatchNorm + LeakyReLU]	$(B, 256, 1)$
Classifier	$(B, 256, 1)$	Flatten + Linear + LeakyReLU + Linear	$(B, 10)$

Table 9: PoseModel: 164,538 parameters

85% accuracy on larger class-balanced training set, 93.5% accuracy on original training set

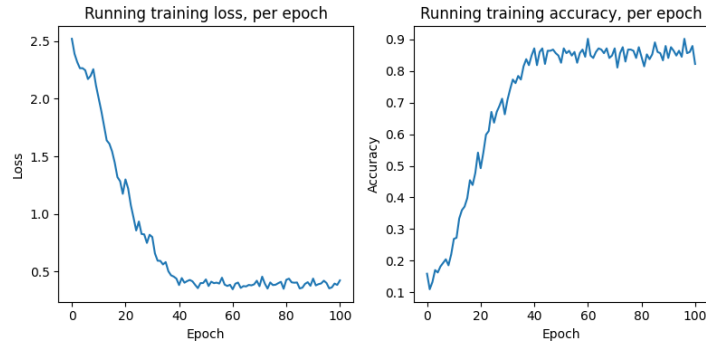


Figure 6: Training loss and accuracy, per epoch

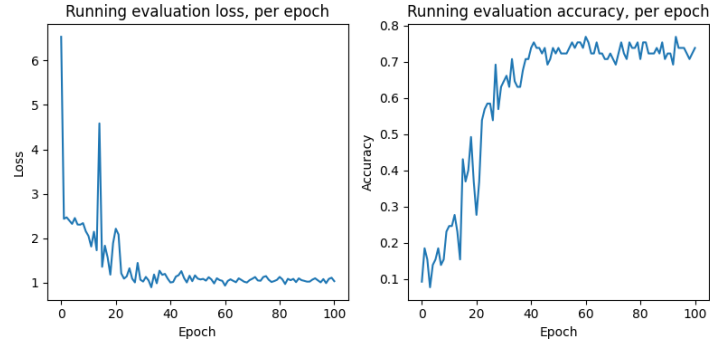


Figure 7: Validation loss and accuracy, per epoch

Our confusion matrix shows nearly all mistakes are on the margin, with the model predicting one up or down of the actual value.

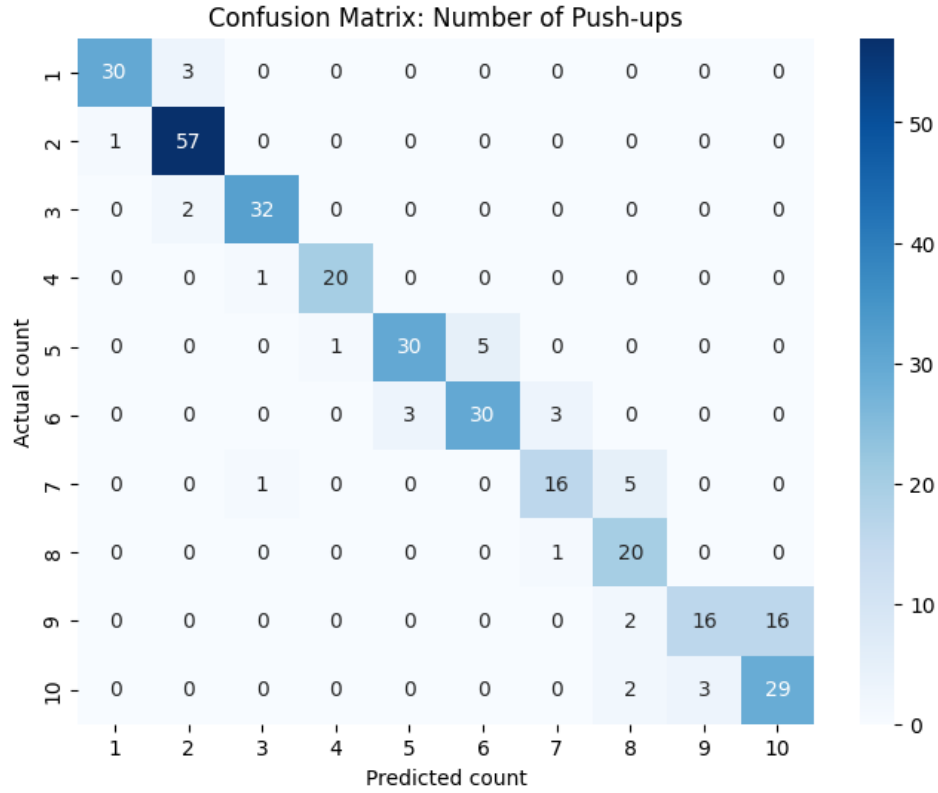


Figure 8: Confusion matrix, predicted vs. actual push-up counts

2.2.1) Small Modificaitons:

- **3D:** exported pose esimations using 3-D coordinates instead of 2-D (still provided by mediapipe). The 3-D locations are in theory more robust to different camera angles.
- **Smoothing** Up to this point, the majority of mistakes tended to be overcounts, i.e the model seeing push-ups where there are none. This is likely due to the model getting confused by small movements. We can smooth the joint coordinates using SavGol smoothing with a moderate window_size (I chose 7) and moderate polynomial degree (I chose 2) to even out small, unhelpful movements. However, too large a smoothing window, and you risk washing out movements you actually want, whereas too large a polynomial degree, and you risk retaining jitters that aren't meaningful.
- **BatchNorm before first linear:** just to make sure that data is centered
- **Opened spatial embedding bottleneck:** Changed Linear(132, 32) to Linear(132, 64), allowing more room for complex spatial representations.
- **Both Average and Max Pooling.** Across several experiments, it seems using Global Average Pooling at the end of the spatial stage improved the model's accuracy on high-count videos, but made it worse on low-count ones (as expressed in the evaluation confusion matrix). Low-count videos are usually shorter and have fewer frames, and the excess padding seems to wash out the real pushup signal from earlier in the video. Switching to Max Pooling helped the model focus on the push-up signal in low count videos, while sacrificing accuracy on the high count ones. The solution seems to be to use both: pool once with the average, once with the max, and then pass both values to the temporal stage. That way, the model has capacity to learn that different combinations of {low/high} {average/max} values tend to yield different counts.
- **Larger classifier input size:** To deal with feeding two pooled inputs instead of one.
- **Remove striding on load.** Previously, we were dealing with motion noise by only loading every 10th frame of each video into each training batch. After frame-poses were smoothed, the model tended to perform worse with any amount of striding, especially on longer videos, where crucial push-up turning points might get lost. For this reason, we load all frames into memory and pad them to 1000 frames as before.
- **Horizontal flipping:** I had forgotten that after transforming videos by duplicating/halving, I could also simply horizontally flip them all, like in (1.2.6). The goal here is to avoid the model overfitting for a certain "handedness" (i.e. the subject slightly leaning to the right or left as they descend).

The final training dataset consisted of 658 videos, with the following class balance:

66, 116, 68, 42, 72, 72, 44, 42, 68, 68]]

Label	1	2	3	4	5	6	7	8	9	10
Count	66	116	68	42	72	72	44	42	68	68

Table 10: (1.2.5) training set labels and counts (329 total)

2.3) Final Model

After applying the above updates, the final model looks like so:

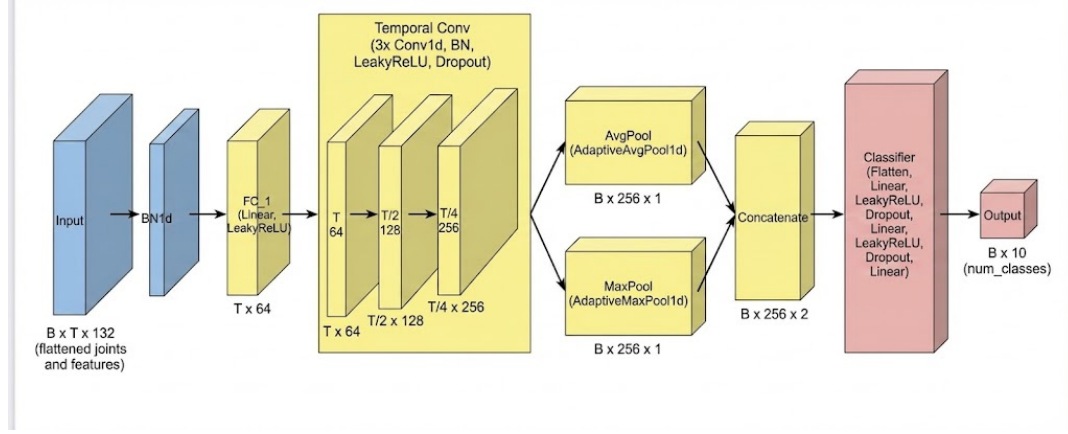


Figure 9: Final Architecture

Stage	Input Shape	Operations	Output Shape
Preprocessing	(B, C, T, H, W)	MediaPipe World Landmark Extraction	$(B, T, 33, 4)$
Data Prep	$(B, T, 33, 4)$	Flatten Joints/Features	$(B, T, 132)$
Spatial Embedding	$(B, T, 132)$	BatchNorm + Linear + LeakyReLU	$(B, T, 64)$
Temporal Processor	$(B, 64, T)$	3x [Conv1d + BatchNorm + LeakyReLU]	$(B, 256, 13)$
Global Pooling	$(B, 256, 13)$	AvgPool + MaxPool (Concatenated)	$(B, 256, 2)$
Classifier	$(B, 256, 2)$	Flatten + 2x (Linear + LeakyReLU) + Linear	$(B, 10)$

Table 11: Final PoseModel: 305,530 parameters

This architecture achieves 94% accuracy on the original training videos, as well as 92% accuracy on our class-balanced set.

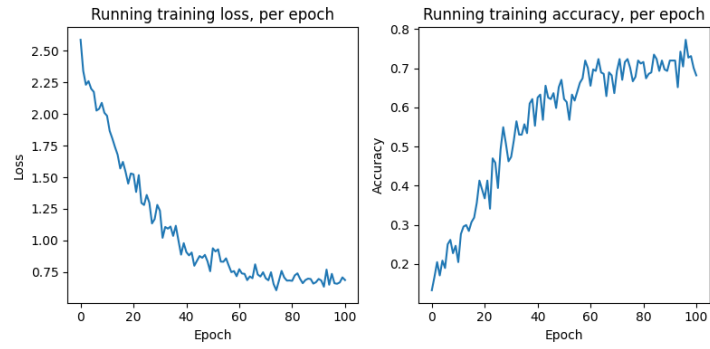


Figure 10: Training loss and accuracy, per epoch

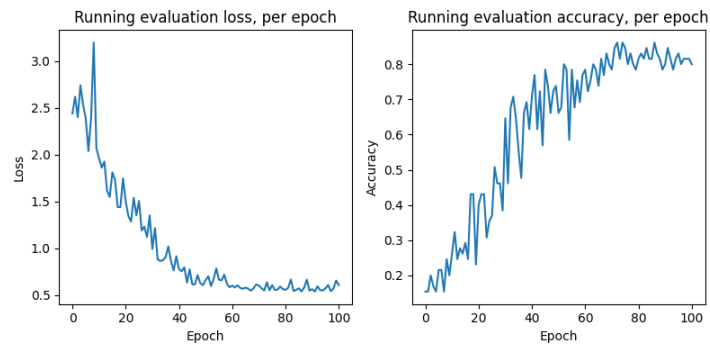


Figure 11: Validation loss and accuracy, per epoch

The confusion matrix (on the class-balanced training set) shows this model still struggles to distinguish between 9 and 10 pushups.

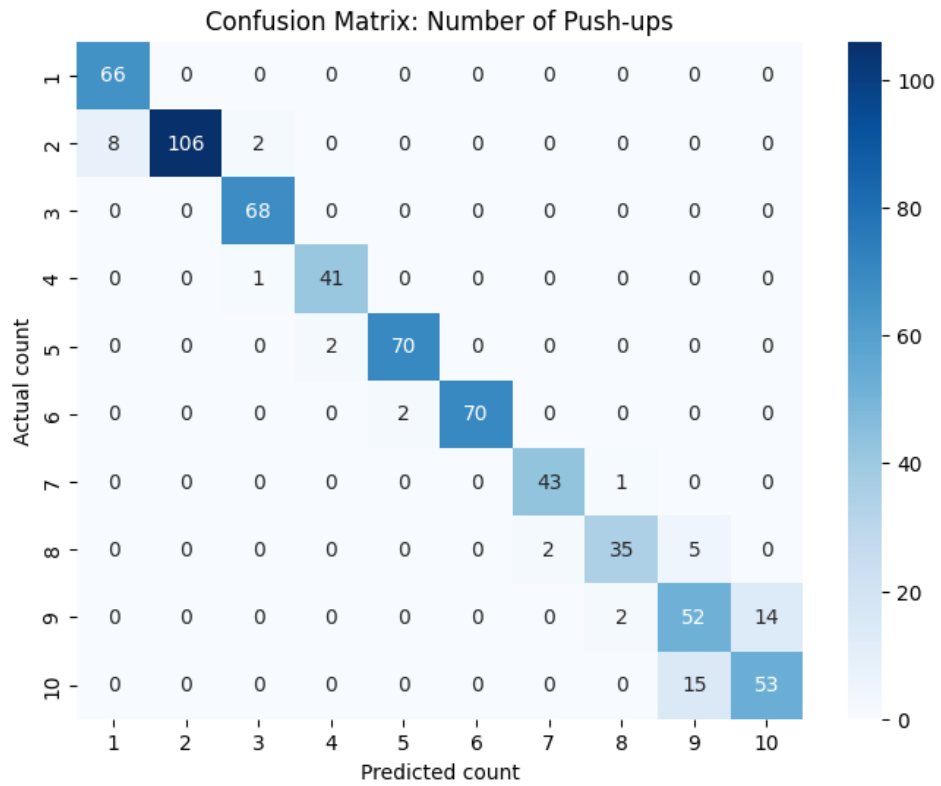


Figure 12: Confusion matrix, predicted vs. actual push-up counts